

✓ Using Google Cloud Shell for Basic Development (Hello World App)

Google Cloud Shell provides a **pre-configured environment** for development without installing anything on your local machine. It allows you to write, run, and deploy cloud applications easily.

◆ Step 1: Open Google Cloud Shell

1. Go to **Google Cloud Console**:
👉 <https://console.cloud.google.com/>
 2. Click on the **Cloud Shell** icon (🟡 Terminal icon) in the top-right corner.
 3. Wait for the Cloud Shell terminal to load. You will get a **Linux-based terminal** with Google Cloud SDK pre-installed.
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◆ Step 2: Create a Project Directory

Once the terminal is ready, create a new directory for your application:

```
mkdir cloudshell-hello
cd cloudshell-hello
```

◆ Step 3: Create the "Hello World" Application

For Python (Flask)

Create a new Python file:

```
touch main.py
nano main.py
```

Add the following **Flask-based Hello World** code:

```
from flask import Flask

app = Flask(__name__)

@app.route('/')
def hello():
    return 'Hello, Cloud Shell!'

if __name__ == '__main__':
    app.run(host='0.0.0.0', port=8080)
```

Save the file (CTRL + X, then Y, then Enter).

◆ Step 4: Install Dependencies

Since Cloud Shell has Python pre-installed, just install Flask:

```
pip install flask
```

◆ Step 5: Run the Application

Start the Flask web server in Cloud Shell:

```
python main.py
```

- Cloud Shell will start running the server on **port 8080**.
 - You will see output like:
 - Running on http://0.0.0.0:8080/
 - Click the **"Web Preview"** button (top-right in Cloud Shell) and select **Preview on Port 8080**.
 - Your browser will open, displaying:
 - Hello, Cloud Shell!
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◆ Step 6: Stop the Application

Press **CTRL + C** in the terminal to stop the server.

✓ Next Steps

- You can now modify the app and add more features.
 - To **deploy it to Google App Engine**, use:
 - `gcloud app deploy`
 - You can also store your project in **GitHub** using:
 - `git init`
 - `git add .`
 - `git commit -m "Initial commit"`
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